Standardised Evaluation for Microbiome Dataset Classifiers

Google Summer of Code 2026 | EMBL-EBI

Prepared: March 23, 2026

Personal Information

Name	Prashant Ranjan
Email	PrashantRanjan0402@gmail.com
GitHub	https://github.com/MrRogueKnight
LinkedIn	https://www.linkedin.com/in/MrRogueKnight/
University	Rajiv Gandhi Institute of Petroleum Technology (RGIPT)
Degree	B.Tech Mathematics & Computing (2024–2028)
Location	Varanasi, India

Executive Summary

This project aims to develop a standardized evaluation and reporting toolkit for Trapiche, a microbiome metadata classification system. Currently, users lack a consistent way to assess model performance and understand how different input components (taxonomic profiles vs text data) influence predictions.

The proposed solution is a modular Python-based evaluation framework that computes standard classification metrics (accuracy, precision, recall, F1-score, AUC-ROC) along with component-aware evaluation for text-only, taxonomy-only, and combined inputs. The system will also include an automated report generator producing human-readable HTML/PDF reports with visualizations such as confusion matrices and comparative performance plots.

The final deliverables will include a reusable evaluation module, automated reporting pipeline, documentation, and a Jupyter notebook walkthrough, enabling researchers to reliably evaluate and interpret microbiome classification models.

Understanding of Trapiche

Trapiche combines two complementary data sources for metadata classification: structured taxonomic profiles derived from microbial composition and unstructured free-text descriptions from project and sample metadata. This dual-input nature makes evaluation non-trivial, as performance depends on how these heterogeneous inputs interact.

The proposed evaluation toolkit will explicitly handle heterogeneous inputs, enable controlled comparisons between modalities, and provide insights into how structured vs unstructured data influence predictions—aligning with Trapiche's design philosophy.

Motivation

Why This Project Matters

• Trust model outputs in real workflows

• Debug and improve models effectively

• Compare results across studies with standardized metrics

• Understand the contribution of text vs taxonomic data

Alignment with My Skills and Passion

• Machine Learning Evaluation using scikit-learn (AUC-ROC, confusion matrix, F1)

• Data visualization for interpretability

• Open source development and documentation (GSSoC 2025)

• Winner, Smart India Hackathon 2025 (ISRO Problem Statement)

Technical Approach

Architecture Overview

```
Trapiche Classifier
(Taxonomic Profiles + Text Descriptions)
|
v
Evaluation Toolkit (This Project)
--------------------------------
| Core Metrics |
| Component-Aware Evaluation |
| Report Generator |
--------------------------------
|
v
Output: Reports | CLI | Notebook | Docs
```

Integration with Trapiche Pipeline

The evaluation toolkit will integrate seamlessly with Trapiche through:

• Accepting Trapiche's standard prediction output formats (JSON/CSV)

• Providing both standalone CLI usage and API-based integration

• Supporting real-time evaluation during model development

• Enabling automated evaluation in CI/CD pipelines

Component-Aware Evaluation Strategy

To systematically measure the contribution of different input modalities, I will implement an ablation-based evaluation framework:

• Text-only evaluation: Using only free-text descriptions

• Taxonomy-only evaluation: Using only microbial composition profiles

• Combined evaluation: Using both inputs together

• Comparison: Δ_combined_vs_text and Δ_combined_vs_taxonomy

Evaluation Reliability & Reproducibility

• Stratified train/test splits for imbalanced datasets

• Cross-validation support (k-fold)

• Seed-controlled experiments for exact reproducibility

• Weighted metrics (macro/micro averaging) and precision-recall curves

• Clear separation of data, evaluation, and reporting layers

Timeline & Milestones

Program Duration: 175 hours (approx. 15-20 hours/week)

Period	Activities	Deliverables
Weeks 1-2	Implement base metric calculation functions	Tested metrics module
Weeks 3-4	Add component-aware evaluation	Text/taxonomy/combined comparison
Weeks 5-6	Write comprehensive unit tests	Test coverage >80% - Mid-term Evaluation
Weeks 7-8	Build HTML report generator	Basic reports with visualizations
Weeks 9-10	Add PDF export and refine	Full reporting suite
Weeks 11-12	Documentation & Jupyter notebook	Complete documentation
Final Weeks	Polish & final submission	Final Evaluation submission

Deliverables

Deliverable	Description
Python Evaluation Module	Complete module with metrics, component evaluation, error analysis
Report Generator	HTML reports with optional PDF export; confusion matrices, performance plots
CLI Interface	Command-line tool for running evaluations
Documentation	Installation guide, API reference, usage examples
Jupyter Notebook	Step-by-step walkthrough with real datasets
Test Suite	Unit tests with >80% coverage, integration tests

Stretch Goals

- Confidence intervals via bootstrap sampling
- Model calibration metrics (Brier score, calibration curves)
- Interactive dashboard using Plotly/Dash
- Docker container for easy deployment

Why Me?

Experience	Relevance
Summer Analytics 2025 (Top 10% of 10,000+)	Built ML models using scikit-learn; focused on metrics
Capstone: Dynamic Pricing Engine	Applied feature engineering, model evaluation
GirlScript Summer of Code 2025	Open source contributor; Git, PR reviews, documentation
DRISHTI: SIH 2025 Winner (ISRO)	Built real-time decision system; systems thinking
Mathematics & Computing Major	Strong analytical foundation

Technical Skills

Category	Technologies
Languages	Python (advanced), SQL, Java, C++
ML & Data	scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Tools	Git, Jupyter, pytest, Docker, CLI development
Concepts	Classification metrics, feature engineering, cross-validation

Post-GSoC Commitment

- Maintain and improve toolkit based on user feedback
- Add features requested by the community
- Integrate the toolkit into Trapiche's main codebase

• Continue contributing to EMBL-EBI open source projects

Appendix: Sample Metrics Output

Example from my Summer Analytics capstone project:

```
Classification Report: Random Forest Classifier
================================================
              precision    recall  f1-score   support
     Class 0       0.87      0.91      0.89       245
     Class 1       0.89      0.84      0.86       267
    accuracy                           0.88       512
   macro avg       0.88      0.88      0.88       512
weighted avg       0.88      0.88      0.88       512
Confusion Matrix:
[[223  22]
 [ 43 224]]
AUC-ROC: 0.94
```

Declaration

• I have read and will adhere to the GSoC 2026 program rules

• I am available to commit 15-20 hours per week during the coding period

• I will communicate regularly with mentors via email, Slack, and scheduled meetings

• I am committed to producing high-quality, well-documented, and tested code

Prepared by: Prashant Ranjan

Date: March 23, 2026